1. Jack has deposited $6,000 in a money market account with a variable interest rate. The account compounds the interest monthly. Jack expects the interest rate to remain at 8% annually for the first 3 months, at 9% annually for the next 3 months, and then back to 8% annually for the next 3 months. Find the total amount in this account after 9 months.
2. You decide to put $12,000 in a money market fund that pays interest at the annual rate of 8.4%, compounding it monthly. You plan to take the money out after one year and pay the income tax on the interest earned. You are in the 15% tax bracket. Find the total amount available to you after taxes.
3. In order to buy a house you want to accumulate a down payment of $15,000 over the next four years. You can do that by putting a certain sum of money in a savings account on the first of every month for the next 48 months. The account credits interest every month at the annual rate of 6%. What is your required monthly deposit?
4. You would like to buy an iPad from your friend who is asking $400 for it. However, you offer to pay for it in 3 monthly installments of $140 each, and you will pay the first $140 after one month. Find the implied *annual* interest rate in your offer.
5. Suppose you have borrowed 72,000 as a mortgage loan on your house. The interest rate is 6%. The bank has calculated the monthly payment to be $515.83. How long will it take you to pay the loan?
6. A father is planning a savings program to put his daughter through college. His daughter is now 13 years sold. She plans to enroll at the university in 5 years, and it should take her 4 years to complete her education. Currently, the cost per year (for everything—food, clothing, tuition, books, transportation, and so forth) is $12,500, but a 5 percent inflation rate in these costs is forecasted. The daughter recently received $7,500 from her grandfather's estate; this money, which is invested in a bank account paying 8 percent interest compounded annually, will be used to help meet the costs of the daughter's education. The rest of the costs will be met by money the father will deposit in the savings account. He will make 6 equal deposits to the account—one in each year from now until his daughter starts college. These deposits will begin today and will also earn 8 percent interest.
   a. What will be the present value of the cost of four years of education at the time the daughter becomes 18?
   b. What will be the value of the $7,500 which the daughter received from her grandfather's estate when she starts college at age 18?
   c. If the father is planning to make the first of 6 deposits today, how large must each deposit be for him to be able to put his daughter through college?
7. Your broker offers to sell you a note for $13,250 that will pay $2,345.05 per year for 10 years. If you buy the note, what rate of interest will you be earning?
8. Assume that you inherited some money. A friend of yours is working as an unpaid intern at a local brokerage firm and her boss is selling some securities which call for four payments, $50 at the end of each of the next 3 years, plus a payment of $1,050 at the end of Year 4. Your friend says she can get you some of these securities at a cost of $900 each. Your money is now invested in a bank that pays an 8 percent nominal interest rate but with quarterly compounding. You regard the securities as being just as safe, and as liquid, as your bank deposit, so your required effective annual rate of return on the securities is the same as that on your bank deposit. You must calculate the value securities to decide whether they are good investment. What is their present value to you?